

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think> CD21 is a receptor for EBV. By modifying B cells to not express CD21, the binding and entry of EBV into the cells would be affected, thus impacting the pathogenesis of EBV. Measles virus, HIV, and human papillomavirus do not primarily use CD21 for entry into cells. </think><answer>A. Epstein-Barr virus (EBV)</answer>

